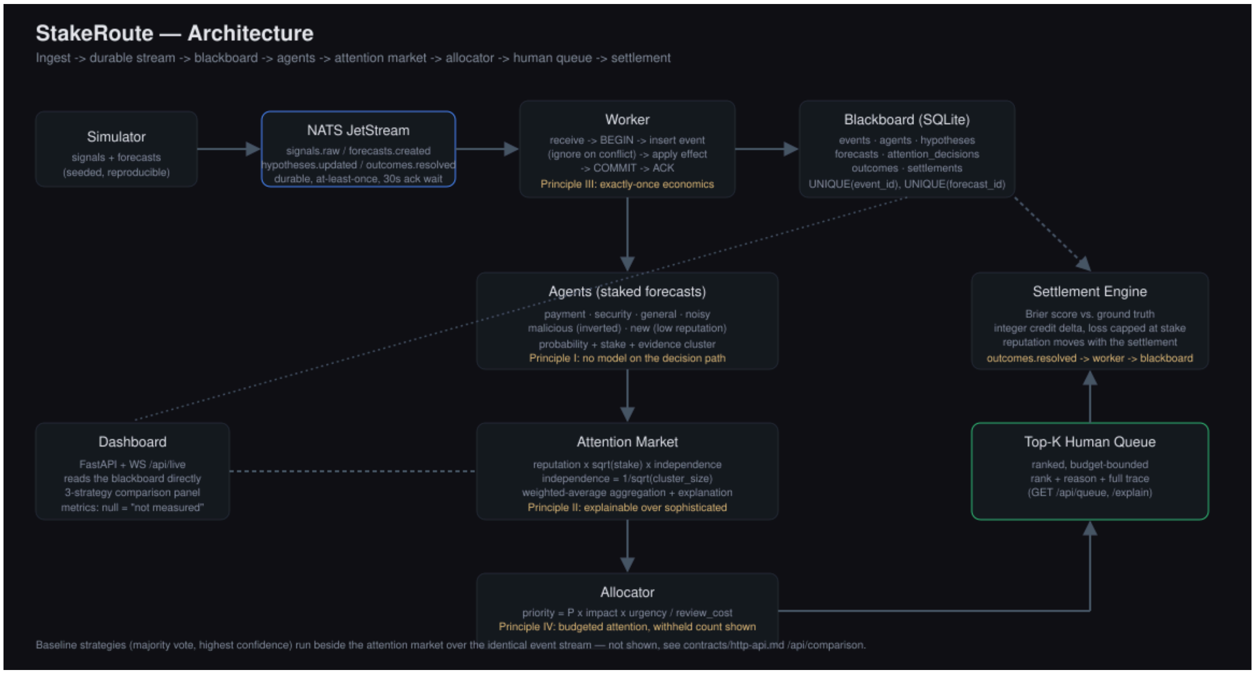
STAKE ROUTE

Design Document · Attention Allocation for Autonomous AI Agents



Demo video: https://drive.google.com/drive/folders/1UhR8bDnBeMHGySNLfoIhpdLGmEKpdbL1?usp=share_link

Executive Summary

StakeRoute is a deterministic attention market for teams running many autonomous AI agents. Agents forecast which incident hypotheses are true, stake scarce credits on those forecasts, and earn or lose reputation after outcomes settle. The system aggregates independent evidence, discounts correlated copies, and sends only the highest-value cases to a limited human review queue. The goal is not to make agents perfectly truthful; it is to allocate scarce human attention better under noise, duplication, and adversarial behavior.

1. Problem & Goals

A company may have dozens of agents monitoring payments, databases, security, and deployments. During an incident they can emit more claims than operators can inspect. Simple voting rewards volume, confidence-only ranking rewards overstatement, and LLM ranking is costly, nondeterministic, and difficult to audit.

- **Primary goal:** rank incident hypotheses so the review budget is spent on the cases with the highest expected operational value.
- **Success criteria:** preserve the correct top result under Sybil and correlated-agent attacks; survive worker failure without losing or duplicating forecasts; explain every score; and sustain bursty event intake.
- **Non-goals:** permissionless identity, autonomous remediation, and proof of a fully strategy-proof market.

2. Architecture & Data Flow

1) Signal ingestion — monitors publish normalized evidence to a JetStream subject.
2) Durable worker — a pull consumer reads events, writes hypotheses and forecasts to SQLite, and acknowledges only after commit.
3) Agent layer — specialized agents submit probability, stake, confidence, and an evidence-group identifier.
4) Attention market — forecasts are weighted by reputation and diminishing-return stake; correlated evidence groups receive a diversity discount.
5) Allocator — deterministic expected-value scoring selects the top-K hypotheses that fit the human review budget.
6) Settlement — verified outcomes update the append-only ledger, available credits, and agent reputation.

3. Decision Mechanism

Forecast influence = reputation × √stake × correlation discount. The square root gives agents more influence for meaningful stake without allowing one wealthy agent to dominate linearly. Forecasts are combined into one probability per hypothesis, then ranked by:

$$\text{priority} = \text{probability} \times \text{impact} \times \text{urgency} \div \text{review cost}$$

This separates belief aggregation from resource allocation: the market estimates what is likely; the allocator decides what deserves attention now. Tie-breaking is stable, so identical inputs always produce the same queue and audit trail.

4. Reliability & Data Model

Every event has a unique ID. The worker commits before ACK; redelivery after a crash is safe because forecasts are protected by uniqueness constraints. Tenant IDs are carried through events and tables to prevent cross-tenant mixing. The ledger is append-only, while settlement is transactional, so credits and reputation cannot diverge halfway through an update.

5. Key Design Decisions

- Deterministic scoring over LLM ranking — cheaper, reproducible, testable, and explainable. LLMs may propose hypotheses, but they are off the final allocation path.
- Weighted probability over a complex prediction-market maker — easier to explain in a five-minute demo and sufficient to test incentives; it sacrifices stronger economic guarantees.
- JetStream plus an idempotent database boundary — provides burst buffering and at-least-once delivery without requiring distributed transactions.
- SQLite for the demo — minimal setup and easy inspection. Production would use Postgres, row-level tenant isolation, and horizontally partitioned consumers.
- Enterprise-attested agent identities — reputation must be costly to reset. A permissionless deployment would need stronger identity and anti-collusion mechanisms.

6. Failure Modes, Scale & Trade-offs

Hidden correlation can still inflate confidence when agents omit or mislabel shared evidence. Reputation learns slowly after distribution shifts. Settlement depends on trustworthy ground truth. SQLite eventually becomes the write bottleneck, and one consumer cannot absorb unlimited bursts. At scale, partition by tenant or incident, move state to Postgres, autoscale consumers on queue lag, and cap per-agent and per-evidence-group exposure.

7. Validation

The test suite covers 53 cases. In the baseline, the true payment incident ranks first. In the Sybil scenario, a false claim wins by raw headcount while StakeRoute keeps the true incident first. In the correlated-agent scenario, repeated copies move probability only within the configured five-point bound. A kill-and-restart test confirms zero lost and zero duplicated forecasts.

Result: operators review fewer, higher-value cases, and every ranking can be reproduced from the event, forecast, ledger, and settlement records.